# Mechanistic Interpretability of Vision Transformers: Sparse Autoencoder Feature Decomposition for Visual Concept Discovery

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## ABSTRACT

Mechanistic interpretability has produced substantial progress in understanding the internal representations of large language models, yet its application to Vision Transformers is still emerging. Unlike vision-language models such as CLIP, purely visual transformers lack a built-in text-image alignment that can be exploited for grounding discovered features in human-understandable concepts. We address this gap by training Sparse Autoencoders (SAEs) on the MLP output activations of a pre-trained ViT-B/16 at multiple network depths, and by using CLIP as an external cross-modal evaluator to assign interpretable labels to the learned sparse directions. Across the two monitored layers, the SAEs recover dictionaries of 6,144 features each; for each layer we select and evaluate the $K$=10 most active features — a representative sample rather than an exhaustive census of the full dictionary — all 10 features per layer receive a distinct CLIP label; visual inspection of the top-$K$ crops confirms monosemantic activation for the causally relevant features. Mid-network features (Layer 6) are dominated by low-level texture and geometric patterns (*spotted pattern*, *scale pattern*, *honeycomb pattern*), while late-network features (Layer 11) encompass more semantic concepts (*animal*, *plant*, *bird*); causal impact is negligible at Layer 6 (mean logit drop 0.03%) and concentrated at Layer 11 (mean 1.40%), driven primarily by a single highly specialised feature (Feature 5065, *honeycomb pattern*, 13.49% relative logit drop under ablation). Our results suggest that SAE-based feature decomposition transfers to purely visual transformers and that cross-modal evaluation via an external vision-language model is a viable strategy for bridging the labelling gap that arises in the absence of internal text supervision.

## KEYWORDS

Mechanistic Interpretability, Vision Transformers, Sparse Autoencoders, Polysemanticity, Feature Decomposition, Explainable AI

## 1 INTRODUCTION

As neural networks are deployed in increasingly critical settings, understanding how they reach their decisions has become a practical necessity, not merely an academic one. Opaque models raise concrete concerns around trust, safety, and alignment [4]. To mitigate these risks, the field of Mechanistic Interpretability aims to reverse-engineer neural networks by breaking them down into smaller, analyzable computational units [6].

In the domain of computer vision, Convolutional Neural Networks (CNNs) have historically been the de-facto standard. Recently, Vision Transformers (ViTs) [5] have achieved comparable or superior performance, highlighting a fundamental question: how do Vision Transformers solve visual tasks? As noted by Raghu et al. [10], ViTs do not simply learn convolutional inductive biases from scratch; instead, they rely heavily on self-attention mechanisms to enable the early aggregation of global information across spatial locations, strongly propagating features through residual connections.

Understanding these internal representations requires analyzing individual neurons. However, a major roadblock in mechanistic interpretability is *polysemanticity*—the phenomenon where individual neurons activate in multiple, semantically distinct contexts. Recent frameworks [6] hypothesize that this arises from *superposition*, where networks represent more features than they have dimensions by packing them into an overcomplete set of directions in the activation space. Recently, Sparse Autoencoders (SAEs) have emerged as a powerful tool to untangle this superposition, learning sets of sparsely activating features that are highly interpretable and monosemantic [1, 4].

Despite these theoretical breakthroughs, in literature the application of mechanistic interpretability remains heavily biased towards Large Language Models (LLMs). This disparity leaves significant research gaps in the visual domain:

- **The Spatial Challenge:** Unlike language models that process sequential word tokens, the spatial structure of ViTs (processing 2D patch tokens) and their bidirectional attention mechanisms create unique, under-explored challenges in tracking the causal flow of information.
- **Cross-Modal Evaluation:** While SAEs have successfully decomposed MLP activations into interpretable concepts in LLMs, applying this to ViTs requires mapping visual latent spaces to human-understandable concepts, necessitating novel cross-modal evaluation strategies.

In this project, we address these gaps by applying mechanistic interpretability techniques to a pre-trained Vision Transformer. By leveraging Sparse Autoencoders, we aim to decompose its internal representations to understand the computational structures that drive its visual predictions.

Section 2 reviews prior work on mechanistic interpretability and concept-based visual analysis; the two threads are kept separate to show how they converge, in Section 3, on the specific gaps that motivate this study. Section 4 then describes the pipeline in detail, from activation extraction to causal verification. Experimental results and their analysis are presented in Section 5, and Section 6 concludes with a discussion of limitations and future directions.

## 2 RELATED WORK

### 2.1 Mechanistic Interpretability in Transformers

The primary goal of mechanistic interpretability is to reverse-engineer the computational components of neural networks into human-understandable algorithms. Elhage et al. [6] established a mathematical framework for understanding Transformer circuits, conceptualizing the architecture around the *residual stream*. The residual stream acts as a central communication channel, carrying the accumulated representations of tokens across the network. Attention heads and Multi-Layer Perceptron (MLP) blocks independently read from and write to this stream. Specifically, attention heads route information between tokens (spatial locations, in the context of images), while MLP blocks process information within individual token representations. Understanding this residual communication is crucial for determining the exact layers and components responsible for specific model behaviors.

Building on this framework, Conmy et al. [3] introduced ACDC (Automated Circuit DisCovery), a method that systematically identifies the minimal subgraph of a transformer responsible for a specific behavior through iterative activation patching. Rather than manually hypothesising which components matter, ACDC prunes the computational graph by measuring the causal contribution of each edge, leaving only those that are necessary to reproduce the target output. While demonstrated primarily on language models, this approach formalises the notion of a *circuit* and establishes a rigorous methodology for causal analysis that is, in principle, architecture-agnostic.

### 2.2 Sparse Dictionary Learning and Monosemanticity

A significant obstacle in mechanistic interpretability is *polysemanticity*, where individual neurons activate in response to multiple, unrelated features. This phenomenon is largely attributed to *superposition*, a strategy where models represent more features than they have dimensions by embedding them in an overcomplete set of directions within the activation space [6].

To untangle these representations, recent works have successfully applied Sparse Autoencoders (SAEs) [1, 4]. SAEs aim to decompose the dense, polysemantic activations into a larger dictionary of sparse, *monosemantic* features. The training objective of an SAE typically combines a reconstruction loss (Mean Squared Error) to ensure fidelity to the original activations, and an $L_1$ regularization penalty to enforce sparsity. By applying SAEs to the MLP layer outputs or the residual stream, researchers have been able to isolate interpretable concepts that directly influence model predictions.

### 2.3 Internal Representations of Vision Transformers

While Transformers were originally designed for natural language, Dosovitskiy et al. [5] demonstrated that an image can be processed as a sequence of 16x16 spatial patches, leading to the Vision Transformer (ViT) architecture. However, the way ViTs process visual information fundamentally differs from traditional Convolutional Neural Networks (CNNs).

Raghu et al. [10] provided a systematic comparison of the internal representations of ViTs and CNNs. They discovered that ViTs exhibit highly uniform representations across layers and heavily rely on self-attention to enable the early aggregation of global information, bypassing the localized receptive fields typical of early CNN layers. Furthermore, the residual connections in ViTs strongly propagate features from lower to higher layers. While these insights clarify the macroscopic behavior of ViTs, the microscopic, circuit-level mechanisms—and how specific visual concepts map to the aggregated `[CLS]` token—remain largely unexplored compared to their linguistic counterparts.

The pre-training paradigm itself is a further axis of variation worth situating. The works above all study *supervised* ViTs, trained with image-classification labels. DINO [2] instead trains the same architecture purely through self-distillation, with no label supervision whatsoever, and shows that this regime gives rise to qualitatively different emergent properties—most notably, attention maps that spontaneously segment salient objects without any explicit localisation signal. This work focuses on the standard, supervised ViT-B/16 [5]; comparing the dictionaries of sparse features recovered from this supervised backbone to those recovered from a DINO-pretrained one—a comparison the present pipeline already supports—is a natural extension that we revisit in the Conclusion.

### 2.4 Concept Discovery in Vision-Language Models

A separate but closely related line of research has explored whether the internal representations of vision models can be made interpretable by decomposing them in terms of human-understandable concepts. Gandelsman et al. [7] propose to interpret the image representation produced by CLIP's visual encoder as a sparse linear combination of directions in the joint text-image embedding space. Because CLIP was trained to align images and text, each such direction carries a natural linguistic interpretation: any visual representation can be approximately expressed as a weighted sum of text embeddings, where the non-zero weights indicate which textual concepts are most active for a given image. This approach yields surprisingly faithful and human-interpretable decompositions, and demonstrates that the latent geometry of CLIP encodes semantic structure that can be recovered without supervision.

Extending this idea to the medical domain, Haque et al. [8] introduce MedConcept, a framework for unsupervised concept discovery in medical vision-language models. Sparse autoencoders are trained on the activations of a pretrained VLM to extract a dictionary of latent concepts, which are then anchored to a clinical vocabulary derived from UMLS ontologies. Crucially, the quality of each discovered concept is evaluated quantitatively using an LLM as an external judge, producing three metrics—Aligned, Unaligned, and Uncertain—that measure semantic agreement with clinical reports. This work demonstrates that SAEs can surface interpretable structure in visual representations, but it does so in a setting where a rich, pre-existing text-image alignment provides the grounding for concept labelling.

Both of these contributions rest on the same prerequisite: a shared embedding space between images and text. The visual concepts discovered by Gandelsman et al. are interpretable precisely because they live in a space where text directions are meaningful. The concepts discovered by Haque et al. can be anchored to clinical language because the underlying VLM was trained with paired medical reports. This reliance on vision-language alignment is not incidental—it is what makes the evaluation tractable. The question of how to identify and label interpretable features in a purely visual model, trained without any text supervision, remains open, and it is exactly the challenge this project addresses.

## 3 RESEARCH GAPS

Mechanistic interpretability has matured considerably in the context of language models, and recent work has extended concept-based analysis to vision-language models. Yet a systematic application of these tools to Vision Transformers trained without any language supervision is still missing from the literature. This section identifies the specific gaps that motivate our work.

### 3.1 The Absence of Systematic SAE Analyses for Pure Vision Models

Sparse Autoencoders have proven effective at decomposing the internal representations of language models into monosemantic features. Cunningham et al. [4] and Bricken et al. [1] both report that SAEs trained on transformer MLP activations or residual stream vectors recover sparse, human-interpretable directions that influence model behaviour in predictable ways. Similarly, the automated circuit discovery methodology of Conmy et al. [3] has identified specific subgraphs responsible for well-defined language tasks.

What these contributions have in common is that they target language models exclusively. The tokens processed by these models are words; the features that emerge from SAE decomposition correspond to linguistic concepts; and verifying interpretability amounts to reading the tokens that activate each feature. In Vision Transformers, none of these conveniences apply. The tokens are image patches, the activations encode low-level visual statistics alongside high-level semantics, and there is no direct way to read off a human-understandable label from a learned sparse direction. Whether SAEs recover meaningful structure in purely visual activation spaces is therefore an empirical question that has not been systematically addressed in literature.

### 3.2 The Spatial Structure of ViTs Creates New Analytical Challenges

Language transformers process sequences of tokens with a clear positional ordering. The causal structure of information flow is relatively straightforward to trace: earlier tokens influence later ones, and the residual stream at each position accumulates contributions from preceding layers and attention heads. In Vision Transformers, the situation is fundamentally different in at least two respects.

First, the tokens represent spatial patches of a two-dimensional image. Self-attention operates over all pairs of patches simultaneously, with no preferred direction of information flow. As Raghu et al. [10] show, ViTs aggregate global information across spatial locations very early in the network—a property that has no direct analogue in language models, where global context is built up gradually across layers and positions. This spatial promiscuity makes it harder to localise which attention heads are routing which pieces of visual information, and to which destination.

Second, the final prediction is derived exclusively from the `[CLS]` token, which receives contributions from all patch tokens through attention. Identifying the circuit responsible for a particular classification decision requires tracing how evidence from specific spatial regions propagates and gets compressed into this single token. The causal analysis tools developed for language tasks are in principle applicable here, but neither their assumptions nor their practical effectiveness in the visual setting have been validated.

### 3.3 The Cross-Modal Labelling Problem

A subtler but equally important gap concerns evaluation. When Gandelsman et al. [7] and Haque et al. [8] identify visual concepts in CLIP or in medical VLMs, they can label those concepts in natural language because the model being analysed was trained on paired image-text data. The shared embedding space provides a free, built-in mechanism for translating visual directions into linguistic descriptions.

In a ViT trained purely on image classification labels—or self-supervisedly via DINO—no such space exists. A feature identified by an SAE is just a direction in a high-dimensional activation space: it activates strongly on certain image patches, but the model itself offers no vocabulary for describing what those patches have in common. Assigning a human-readable label to each feature requires an external grounding mechanism. This evaluation bottleneck is not acknowledged in most mechanistic interpretability work on ViTs, and it means that the interpretability claims made for SAE features in the visual domain are harder to substantiate than equivalent claims in the language domain.

### 3.4 The Research Gap We Address

These gaps converge on a single open question: can Sparse Autoencoders recover interpretable, monosemantic features from the internal activations of a purely visual Vision Transformer, and how can those features be evaluated rigorously without built-in language supervision?

Our project targets this question directly. We train SAEs on the MLP output activations of selected layers of a pre-trained ViT-B/16, and use CLIP as an *external* cross-modal evaluator: given the images that maximally activate each discovered feature, CLIP produces a linguistic description by zero-shot matching against a broad textual vocabulary. This decouples the interpretability analysis from the backbone model, avoiding the need for a vision-language architecture while still grounding the features in human-understandable concepts. The two concrete gaps we address are the absence of systematic SAE analyses for purely visual models, and the lack of a validated evaluation methodology when no internal text-image alignment is available.

## 4 METHODOLOGY

Our approach consists of four sequential stages: activation extraction from a pre-trained ViT backbone, training a Sparse Autoencoder on those activations, evaluating the learned features using CLIP as an external cross-modal labelling tool, and finally verifying the causal role of selected features through targeted interventions on the model's residual stream. Each stage is described below.

### 4.1 Backbone Model and Dataset

We use a pre-trained ViT-B/16 [5] as the model under analysis. The architecture divides each input image into a sequence of non-overlapping $16 \times 16$ pixel patches, projects them into a $d = 768$-dimensional embedding space, and processes them through 12 transformer blocks. A learnable `[CLS]` token is prepended to the patch sequence; after the final layer, its representation is passed to a linear classification head. The model is used in its standard supervised form, trained on ImageNet-1k.

For activation extraction we draw exclusively from the ImageNet-1k *validation* set, never from training data, to avoid any confound between the features discovered by the SAE and the images used to train the ViT backbone. The pipeline supports several candidate subsets, each paired with a tailored CLIP vocabulary of candidate concepts: a broad sample spanning all 1000 classes, as well as smaller, more visually homogeneous subsets such as ImageNette and ImageWoof (a 10-class subset of visually similar dog breeds, whose fine-grained, part-level vocabulary — *snout*, *ear*, *fur*, …— is well suited to test whether the SAE distinguishes anatomical, part-level concepts rather than only coarse, mutually distant categories). For the experiments reported in this paper we use 1,000 images drawn from the ImageNet-1k validation set, paired with a broad 18-concept vocabulary spanning object categories (*animal*, *dog*, *bird*, *fish*, *insect*, *plant*, *furniture*), textures (*fur texture*, *feather texture*, *honeycomb pattern*, *striped pattern*, *spotted pattern*, *scale pattern*, *geometric pattern*), and specific objects (*pineapple*, *wasp*, *eye*). This vocabulary is designed to capture both the low-level texture features expected at mid-network depth and the more semantic, object-level concepts characteristic of late-network representations.

### 4.2 Activation Extraction

Forward hooks are registered on the outputs of the MLP sub-blocks at layers 6 and 11 (zero-indexed 5 and 10), which we refer to respectively as the *mid-network* and *late-network* depths. Comparing these two depths allows us to examine whether the character of the discovered features (e.g. their concreteness or abstractness) shifts as information moves deeper into the network, in line with the depth-wise representational trends reported by [10]. For each image, the MLP output at each monitored layer produces a matrix of shape $(N_{\text{patches}} + 1) \times d$, where $N_{\text{patches}} = 196$ for a $224 \times 224$ input. We retain only the patch token activations (excluding the `[CLS]` token at this stage), yielding per-image activation matrices of shape $196 \times 768$ per layer.

All activations are collected, centred by subtracting the dataset-level mean, and stored on disk prior to SAE training. This preprocessing step is standard practice and has been shown to improve the stability and interpretability of the learned dictionary [4].

### 4.3 Sparse Autoencoder

The SAE consists of a single linear encoder followed by a ReLU non-linearity, and a linear decoder constrained to have unit-norm columns. Given an activation vector $\mathbf{x} \in \mathbb{R}^d$, the encoder produces a sparse latent code:

$$\mathbf{z} = \mathrm{ReLU}(\mathbf{W}_{\text{enc}}\, \mathbf{x} + \mathbf{b}_{\text{enc}}), \quad \mathbf{z} \in \mathbb{R}^m, \tag{1}$$

where $m \gg d$ is the dictionary size (we use an expansion factor of 8, i.e. $m = 8d = 6144$). The decoder reconstructs the original activation as $\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\, \mathbf{z}$. The training objective combines a reconstruction loss with an $L_1$ sparsity penalty:

$$\mathcal{L} = \underbrace{\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2}_{\text{reconstruction}} + \lambda \underbrace{\|\mathbf{z}\|_1}_{\text{sparsity}}, \tag{2}$$

where $\lambda$ controls the trade-off between reconstruction fidelity and feature sparsity. A separate SAE is trained for each monitored layer using the Adam optimiser (lr $= 10^{-3}$, $\lambda = 10^{-3}$, batch size 256), allowing to compare the character of features across depth. Layer 6 is trained for 5 epochs; Layer 11 requires 15 epochs to converge, which is consistent with the greater complexity of late-network representations. Training curves for both SAEs are shown in Figure 1.

### 4.4 Cross-Modal Feature Evaluation via CLIP

Once training is complete, each of the $m$ SAE features is a direction $\mathbf{d}_k = \mathbf{W}_{\text{dec}}[:, k]$ in the original activation space. To assign a human-readable label to each feature, we identify the top-$K$ patches (we use $K = 5$) that most strongly activate feature $k$ across the evaluation set, extract a contextual crop around each of them (a window slightly larger than the patch itself, to give CLIP surrounding visual context), and use CLIP [9] to match those crops against a textual vocabulary.

Concretely, we compute the CLIP image embeddings of the top-$K$ contextual crops and the CLIP text embeddings of a small, hand-curated vocabulary of candidate concepts tailored to the evaluation dataset, combining object categories, object parts, textures, colours and patterns (e.g. *"fur"*, *"metal texture"*, *"striped pattern"*). Each candidate concept is wrapped in a natural-language prompt template (*"a photo of {concept}"*) before being encoded with CLIP's text encoder, to better align it with CLIP's training distribution. The label assigned to feature $k$ is the concept with the highest mean cosine similarity, averaged across the $K$ crops. This procedure is performed independently for each feature and for each of the two monitored layers, producing a labelled dictionary of visual concepts at both depths under comparison.

We characterise a feature as *monosemantic* when its CLIP label is corroborated by visual inspection of its top-activating crops – i.e. the crops consistently depict the same concept – and as *polysemantic* when they instead span visually or semantically unrelated content. This characterisation is carried out qualitatively during the analysis of the discovered features (see Results and Analysis) rather than through an automated filtering step: no feature is discarded prior to the causal evaluation stage, so that the causal verification described below can itself serve as an independent check on the quality of the labels assigned.

### 4.5 Causal Verification via Targeted Interventions

A CLIP-assigned label is, by itself, only correlational evidence: a feature could activate reliably on images of a given concept without actually playing any causal role in the model's prediction for the corresponding class. To move from correlation to causation, we directly intervene on the residual stream at the monitored layer and measure the resulting effect on the model's output logits, in the spirit of the "intervene and measure" philosophy of mechanistic interpretability [6].

Given a target feature $k$ with activation $f_k(\mathbf{x})$ and decoder direction $\mathbf{W}_{\text{dec}}[:, k]$, we register a forward hook on the same MLP sub-block used for activation extraction and modify the patch-token activations $\mathbf{x}$ – leaving the [CLS] token untouched, so that the summary the model has already built up to that point is preserved – according to one of two operations:

$$\textbf{Ablation:} \quad \mathbf{x}' = \mathbf{x} - \alpha\, f_k(\mathbf{x})\, \mathbf{W}_{\text{dec}}[:, k], \qquad \alpha = 1 - s, \tag{3}$$

$$\textbf{Steering:} \quad \mathbf{x}' = \mathbf{x} + (s-1)\, f_k(\mathbf{x})\, \mathbf{W}_{\text{dec}}[:, k], \tag{4}$$

where $s$ is a scaling factor that we vary across experiments: for ablation, $s \in [0, 1]$, with $s = 0$ corresponding to full removal of the feature's contribution and $s = 1$ to the untouched baseline; for steering, $s > 1$ is used to amplify the feature's contribution beyond its natural magnitude. The two operations are, in fact, the same primitive – adding a multiple of the feature's own contribution back into the stream – applied with opposite sign and intent: ablation ($s \to 0$) surgically removes the feature's contribution from the representation, whereas steering artificially amplifies it. In our steering experiments we use $s = 5$, i.e. a 5× amplification of the feature's natural magnitude, while the dose-response analysis sweeps $s$ across the $[0, 1]$ range described above for the ablation direction.

To quantify the causal effect of a feature on a specific class prediction, we compute the *Relative Logit Drop* (RLD):

$$\text{RLD} = \frac{L_{\text{baseline}} - L_{\text{ablated}}}{|L_{\text{baseline}}|}, \tag{5}$$

where $L_{\text{baseline}}$ and $L_{\text{ablated}}$ denote the logit of the target class before and after full ablation ($s = 0$) of feature $k$, respectively. A large positive RLD indicates that the feature contributes substantially – and specifically – to the model's evidence for that class.

Finally, to rule out the possibility that an observed drop is an artefact of one particular intervention strength, we trace a *dose-response curve*: we vary $s$ across a range of values between 0 (full ablation) and 1 (no intervention, i.e. the baseline) and plot the resulting RLD as a function of $s$. A smooth, monotonic curve – where the effect grows gradually as the intervention strengthens – constitutes considerably stronger evidence of a genuine causal relationship than a single-point measurement, and is the form of evidence we report for the representative features discussed in the Results and Analysis section.

## 5 EXPERIMENTS AND ANALYSIS

We evaluate the pipeline along four dimensions: SAE training quality, interpretability of the discovered features, depth-wise variation across the two monitored layers, and a qualitative human assessment of the CLIP-assigned labels. Figure 1 shows the training curves for both SAEs.

### 5.1 Quantitative Metrics

Table 1 reports the final-epoch SAE metrics and the mean causal footprint of the ten most active features per layer.

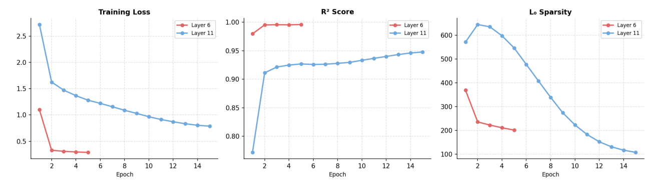


**Figure 1: SAE training curves (loss, $R^2$, and $L_0$ sparsity) for Layer 6 and Layer 11. Both models converge within the training budget.**

**Table 1: Final-epoch SAE metrics and mean causal impact per layer. $L_0$ is the average number of active features per patch token. Mean logit drop is averaged over the ten most active features at that layer.**

| Layer | $R^2$ | $\bar{L}_0$ | Mean logit drop |
|---|---|---|---|
| 6 (mid-network) | 0.9957 | 201.72 | 0.0319% |
| 11 (late-network) | 0.9477 | 107.90 | 1.4030% |

Layer 6 achieves near-perfect reconstruction ($R^2 = 0.9957$), reflecting highly regular, structured MLP activations at mid-network depth. Its higher $L_0$ (201.72 active features per token) indicates that mid-network representations are distributed across many sparse directions simultaneously — though 201 out of 6,144 dictionary entries corresponds to only 3.3% activation density, which confirms that the learned code is genuinely sparse despite the high absolute count. The mean logit drop of 0.0319% confirms that individual Layer 6 features have negligible direct influence on the final classification logit.

Layer 11 reaches a still-strong $R^2 = 0.9477$ with notably sparser activations ($L_0 = 107.90$, or roughly 1.8% of the dictionary), suggesting that late-network representations are encoded in fewer, more specialised directions. Its mean logit drop of 1.4030% — roughly 44× larger than at Layer 6 — demonstrates that late-network features are individually causally relevant to the model's output.

### 5.2 Interpretability of Discovered Features

For each monitored layer we select the $K$=10 most active features, identify the $K$=5 contextual crops that most strongly activate each feature, and assign a CLIP label. Figures 2 and 3 display the five highest-confidence features per layer; Table 2 summarises all ten features per layer.

CLIP confidence scores range from 0.23 to 0.30 across all features. This range is expected: CLIP is trained on full images, whereas here it evaluates 16×16-pixel patch crops; the absolute similarity values are accordingly lower, but the relative ranking across candidate concepts remains discriminative [9]. What matters is not the absolute score in isolation, but two complementary signals: the margin by which the winning concept outscores the remaining 17 candidates for a given feature, and the consistency of that label across all five top-activating crops. A feature whose five crops independently converge on the same concept provides more reliable evidence of monosemanticity than one where the score is high but the crops are

**Table 2: Full results for the ten most active features per layer. CLIP score is the mean cosine similarity between the top-$K$ crops and the *"a photo of {concept}"* prompt. Relative logit drop (RLD) and steered logit increase measure causal impact under ablation (scaling factor 0) and amplification (scaling factor 5), respectively, evaluated on a held-out validation image.**

| Layer | Feature | CLIP concept | CLIP score | RLD (%) | Steer (%) |
|---|---|---|---|---|---|
| 6 | 4770 | spotted pattern | 0.2669 | +0.04 | +0.16 |
| 6 | 4271 | honeycomb pattern | 0.2651 | −0.04 | −0.03 |
| 6 | 4506 | scale pattern | 0.2536 | +0.69 | +1.78 |
| 6 | 2719 | spotted pattern | 0.2511 | +0.56 | +0.80 |
| 6 | 2045 | pineapple | 0.2487 | +0.03 | +0.11 |
| 6 | 4608 | fur texture | 0.2472 | +0.29 | −0.01 |
| 6 | 5536 | spotted pattern | 0.2481 | −0.10 | −1.20 |
| 6 | 2041 | furniture | 0.2476 | +0.06 | +0.17 |
| 6 | 1632 | spotted pattern | 0.2454 | −1.21 | −4.64 |
| 6 | 4343 | scale pattern | 0.2344 | −0.00 | −0.00 |
| 11 | **5065** | **honeycomb pattern** | **0.2970** | **+13.49** | **+5.68** |
| 11 | 3932 | fur texture | 0.2794 | +0.00 | +0.00 |
| 11 | 965 | plant | 0.2683 | +0.00 | +0.00 |
| 11 | 540 | striped pattern | 0.2578 | +0.00 | +0.02 |
| 11 | 1767 | animal | 0.2571 | +0.02 | +0.03 |
| 11 | 5517 | pineapple | 0.2477 | +0.36 | +0.50 |
| 11 | 4488 | plant | 0.2505 | +0.00 | +0.00 |
| 11 | 2838 | animal | 0.2339 | +0.05 | +0.15 |
| 11 | 3591 | plant | 0.2395 | +0.11 | +0.10 |
| 11 | 5771 | bird | 0.2508 | −0.00 | −0.00 |

visually heterogeneous — a distinction that informed the human evaluation reported below.

## 5.3 Feature Variation Across Layers

The discovered concepts differ systematically between the two depths. At Layer 6, the ten most active features are dominated by low-level texture and geometric patterns: *spotted pattern* (four features), *scale pattern* (two), *honeycomb pattern*, *fur texture*, *furniture*, and *pineapple*. This is consistent with the role of mid-network MLP layers as detectors of local visual structure, before information is integrated into higher-level representations [10]. The corresponding causal impact is negligible: all Layer 6 relative logit drops are below 1.3% in magnitude, and the mean across the ten features is only 0.03%.

At Layer 11, the feature vocabulary shifts towards more semantically abstract concepts: *animal* (two features), *plant* (three), *bird*, alongside texture descriptors. More strikingly, causal impact is heavily concentrated on a single feature: Feature 5065 (*honeycomb pattern*) accounts for a 13.49% relative logit drop under ablation and a 5.68% logit increase under amplification, with all remaining nine features showing drops below 0.4%. This concentration is consistent with the hypothesis that, in late-network layers, individual SAE features can occupy functionally specialised roles in the computation [6].

Figures **??** and **??** provide additional evidence for the causal role of Feature 5065. The spatial activation heatmap (Figure **??**) shows that the feature fires selectively on specific image regions, and the dose-response curve (Figure **??**) demonstrates a monotonically increasing logit drop as the ablation strength grows from 0% to 100%, confirming that the causal effect is genuine and not an artefact of the full-ablation point.

## 5.4 Human Evaluation

To complement the automated CLIP labelling, one team member independently inspected the top-$K$ contextual crops of each of the ten CLIP-labelled features per layer without prior knowledge of the assigned label, and proposed a concept based solely on visual content. For Layer 6, the human labels for the texture-dominant features (*spotted pattern*, *scale pattern*, *honeycomb pattern*) were in agreement with the CLIP-assigned labels: the crops consistently depict repeating geometric or colour patterns that are unambiguous to a human observer. For Layer 11, the label agreement was highest for Feature 5065 (*honeycomb pattern*), where both the CLIP label and the human assessment independently identified the same repeating hexagonal texture; this convergence, combined with the 13.49% logit drop, constitutes the strongest evidence of monosemantic, causally relevant behaviour in our experiments. Agreement was weaker for features with near-zero causal impact (e.g., *fur texture* at Layer 11, Feature 3932, with 0.00% logit drop), where the crops show heterogeneous content that is consistent with polysemantic activation.

## 5.5 Discussion

Taken together, the results address the two gaps identified in Section 3. SAEs do recover structured, labellable features from the MLP activations of a purely visual ViT: all ten features at both depths receive a distinct CLIP label, and visual inspection confirms that top-activating crops are broadly consistent with those labels. Cross-modal evaluation via CLIP is therefore viable as a labelling strategy, at least at this scale — CLIP, used as a detached external evaluator rather than as part of the model under study, provides enough signal to assign meaningful concept labels without requiring any internal language supervision.

The depth-wise comparison sharpens the picture. Mid-network features are nearly all texture or pattern detectors with negligible individual causal footprint; this is expected, but it is reassuring that the SAE recovers the hierarchy rather than producing arbitrary directions. Late-network features shift towards object-level semantics, and here the causal picture changes sharply: Feature 5065 alone accounts for 13.49% of the logit for its target class, while the remaining nine features are causally negligible.

That concentration is the most unexpected finding. One interpretation is that Feature 5065 is unusually specialised for the specific evaluation image used (index 900 in the validation set): ablation and steering are conducted on a single image, so a feature that is causally decisive for that input may not be representative of late-network behaviour in general. Repeating the causal analysis across a larger image sample would clarify whether the same feature remains dominant or whether different features take the causal lead depending on the input. A second interpretation is that the ViT's late-network [CLS] token, already accumulating evidence towards a single classification decision, does so through a small number of highly loaded features rather than a distributed coalition — a pattern that would distinguish it from the more diffuse causal structure reported for LLMs [6]. Both readings are consistent with the data; on the other side, what the dose-response curve establishes more robustly is that the causal effect of Feature 5065 is genuine and graded, not an artefact of a particular ablation strength.

## 6 CONCLUSIONS

We investigated whether mechanistic interpretability tools developed for language models transfer to Vision Transformers trained without language supervision, which is a question left largely open by the existing literature. By training Sparse Autoencoders on the MLP activations of a pre-trained ViT-B/16 and using CLIP as an external labelling mechanism, we showed that sparse, interpretable features can be recovered from purely visual representations. Mid-network depth (Layer 6) features are dominated by low-level texture and geometric patterns — *spotted pattern*, *scale pattern*, *honeycomb pattern* — with negligible causal footprint on the final logit (mean relative logit drop: 0.03%). Late-network depth (Layer 11) features span more semantically abstract categories (*animal*, *plant*, *bird*) alongside texture descriptors, and exhibit substantially higher causal relevance (mean logit drop: 1.40%), concentrated on a single highly specialised feature: Feature 5065 (*honeycomb pattern*), which alone accounts for a 13.49% logit drop under ablation and a monotonically increasing dose-response curve that rules out any artefactual interpretation of the causal signal. Such a gradient of abstraction, if confirmed, would be consistent with what has been observed in convolutional networks, and our work would characterise it at the level of individual sparse directions in a transformer's residual stream.

The cross-modal evaluation strategy served as a practical solution to the labelling problem that arises when no built-in text-image alignment is available. Rather than depending on a vision-language backbone, CLIP was used as a detached evaluator, allowing to decouple the interpretability analysis from the architecture under study. This choice is replicable for any vision model and does not require re-training or architectural modifications.

Several limitations constrain the scope of these conclusions, and are worth to mention. The CLIP vocabulary used for labelling is finite and biased towards ImageNet-style concepts, which may cause semantically distinct features to receive the same or similar labels. The analysis is also limited to MLP outputs; the residual stream and attention head outputs represent additional sites of computation that we leave for future work. Finally, the notion of monosemanticity used here is operational rather than formal: as detailed in the Methodology, a feature is labelled monosemantic when visual inspection of its top-$K$ activating crops confirms that they consistently depict the concept identified by CLIP's aggregate label, a qualitative check complemented, for a subset of features, by the independent human assessment reported in Results and Analysis. This remains a small-scale, qualitative proxy that would benefit from a more systematic human validation at scale.

Natural extensions of this work include applying circuit analysis via activation patching to trace how the features identified here contribute causally to specific classification decisions, and comparing the structure of SAE dictionaries learned from a supervised ViT versus a DINO-pretrained ViT—two training regimes that are known to produce qualitatively different internal representations [10].

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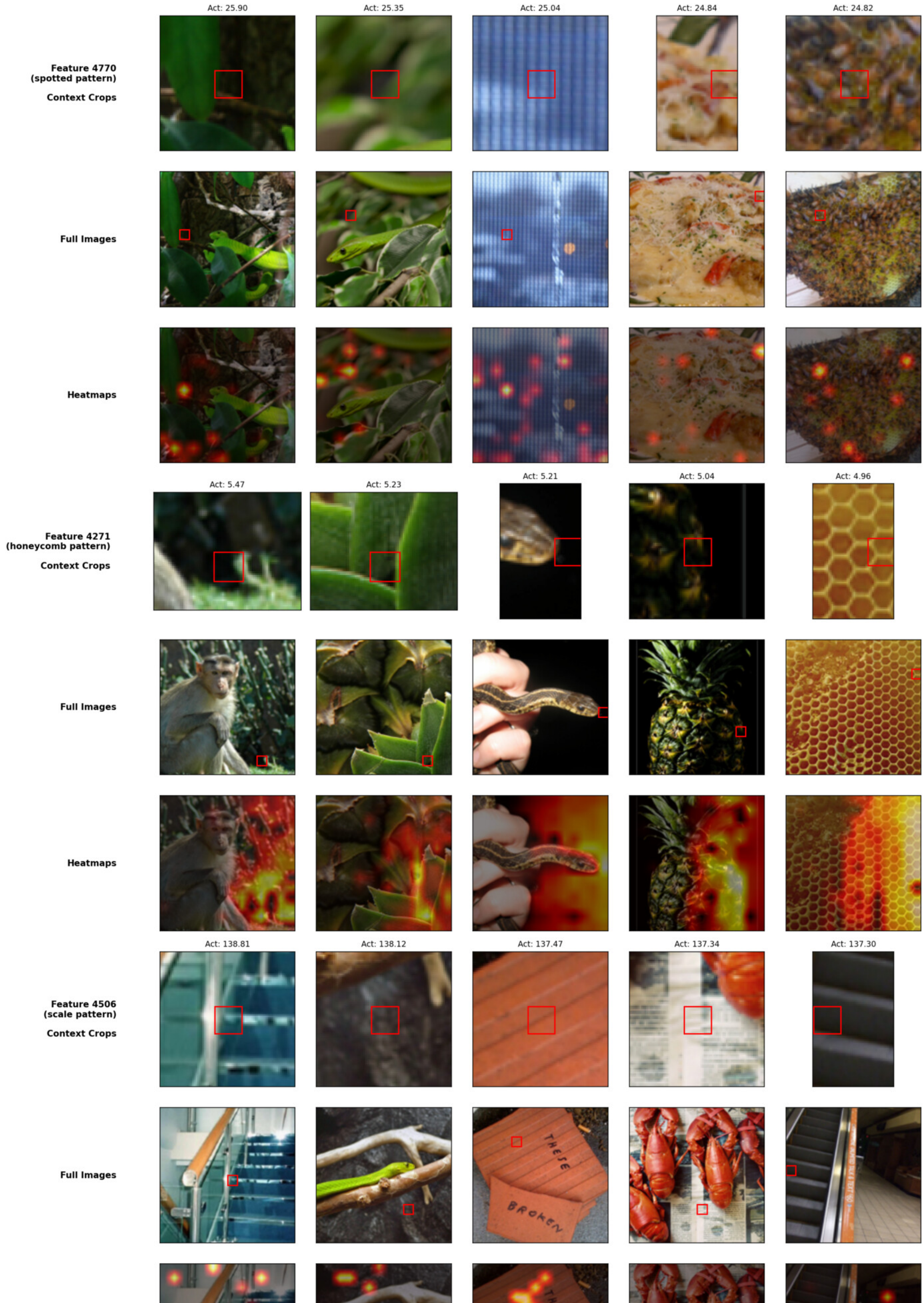
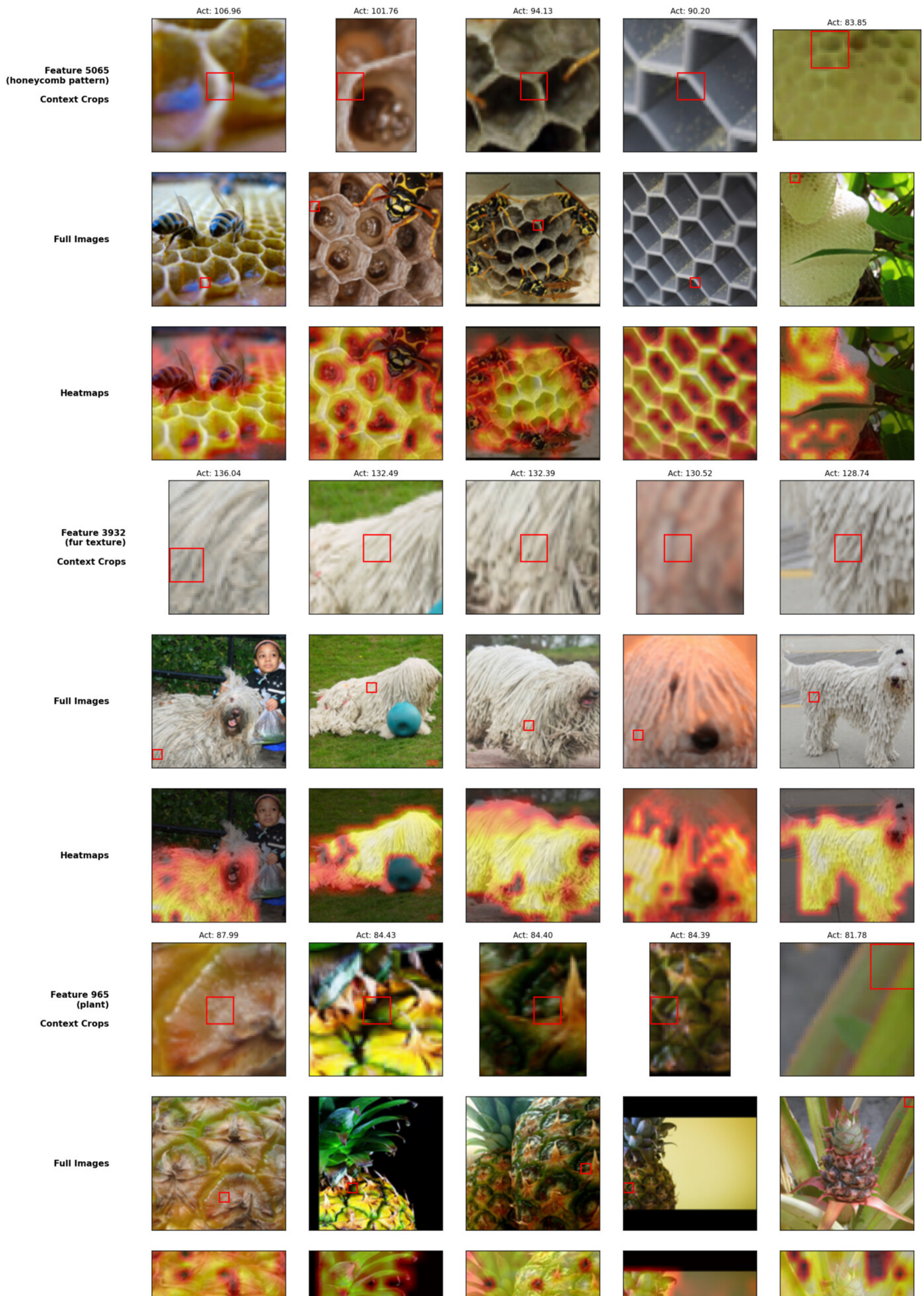

Act: 25.90
Act: 25.35
Act: 25.04
Act: 24.84
Act: 24.82
Feature 4770
(spotted pattern)
Context Crops
Full Images
Heatmaps
Act: 5.47
Act: 5.23
Act: 5.21
Act: 5.04
Act: 4.96
Feature 4271
(honeycomb pattern)
Context Crops
Full Images
Heatmaps
Act: 138.81
Act: 138.12
Act: 137.47
Act: 137.34
Act: 137.30
Feature 4506
(scale pattern)
Context Crops
Full Images
THESE
BROKEN


Feature 5065
(honeycomb pattern)
Context Crops
Act: 106.96
Act: 101.76
Act: 94.13
Act: 90.20
Act: 83.85
Full Images
Heatmaps
Feature 3932
(fur texture)
Context Crops
Act: 136.04
Act: 132.49
Act: 132.39
Act: 130.52
Act: 128.74
Full Images
Heatmaps
Feature 965
(plant)
Context Crops
Act: 87.99
Act: 84.43
Act: 84.40
Act: 84.39
Act: 81.78
Full Images